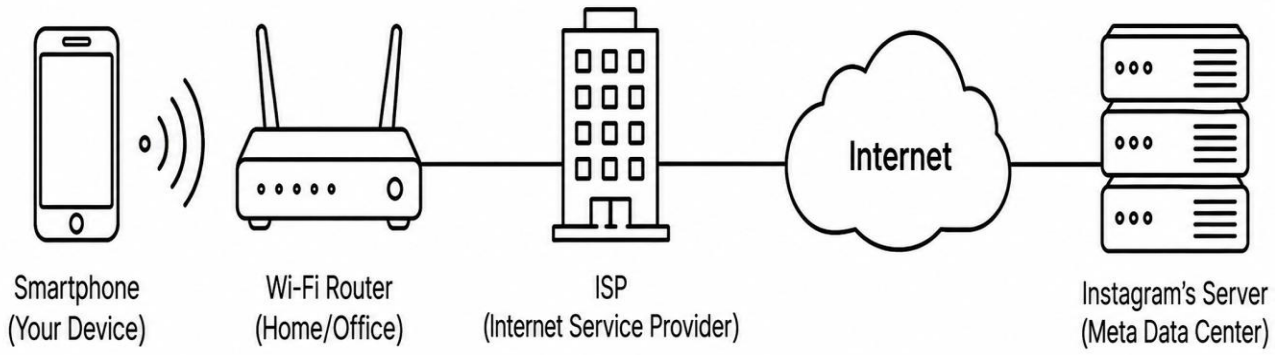
# Networking technologies

**Tasks :-**

**1.Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.**

Ans :-



**2.List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.**

Ans :-

| Item | Example | Meaning |
|---|---|---|
| **IP Address** | `10.79.173.31` | The unique address assigned to your device on the local network. |
| **Default Gateway** | `10.79.173.136` | Usually your Wi-Fi router; it forwards traffic from your local network to other networks/Internet. |

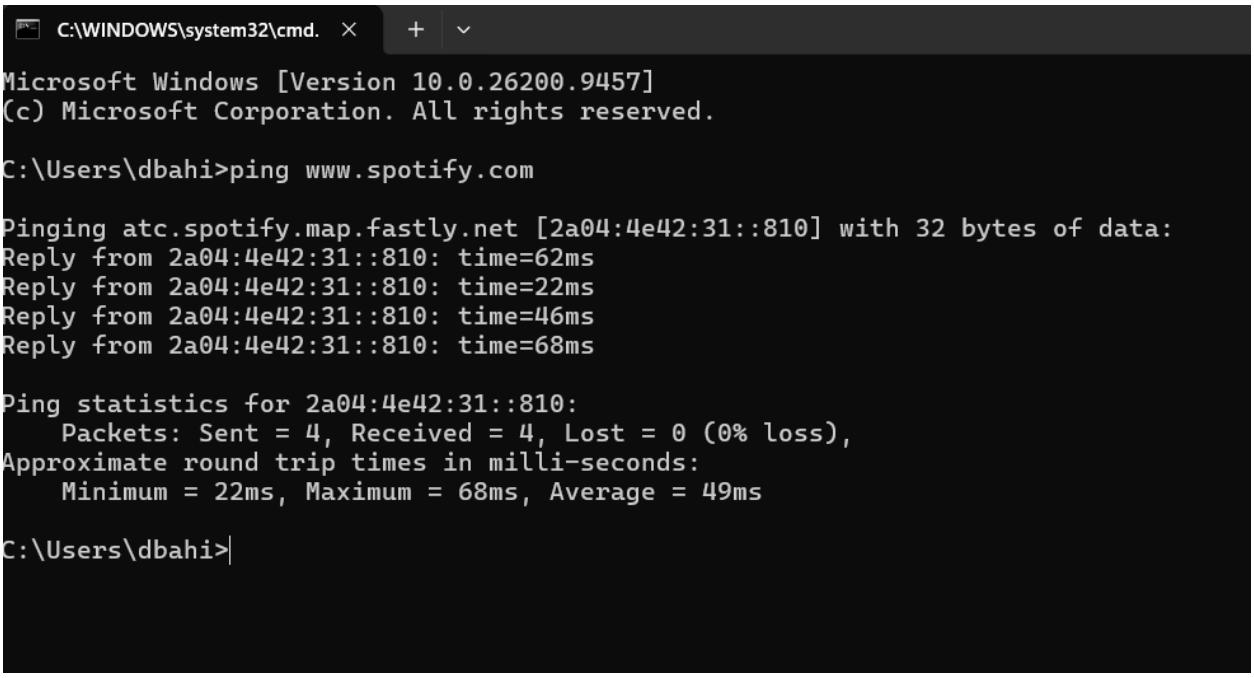
**3.Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant.<br><br>Hint: Think about what happens when you search for a restaurant and place an order.**

Ans :-

| Step | What happens | Protocol |
|---|---|---|
| 1. Open Zomato | Your device connects to the Internet through Wi-Fi/mobile network. | TCP/IP |
| 2. DNS Lookup | Your device asks a DNS server to convert Zomato's domain name into an IP address. | DNS |
| 3. Server Connection | Your device establishes a reliable connection with Zomato's server. | TCP |
| 4. Restaurant Search | Your device sends a request for restaurants/menu information. | HTTP/HTTPS |
| 5. Response | Zomato's server sends restaurant names, menus, prices, etc. back to your device. | HTTPS/TCP/IP |
| 6. Place Order | You select food and place the order; your device sends the order details to the server. | HTTPS/TCP/IP |
| 7. Order Processing | Zomato processes the order and communicates it to the restaurant/payment system. | HTTPS |
| 8. Confirmation | The server sends order confirmation and order details back to your device. | HTTPS/TCP/IP |

**4.Use the 'ping' command to check the connection between your computer and www.spotify.com, then explain what the results mean for network troubleshooting.**

Ans :-




➢ **Troubleshooting :-**

| Result | Possible Meaning |
|---|---|
| `0% loss` | Connection is responding normally |
| High `time` | High network latency |
| Packet loss | Possible Wi-Fi, router, ISP, or network congestion |
| `Request timed out` | Destination may not respond to ICMP, or there may be a connectivity problem |
| `Ping request could not find host` | DNS/name-resolution problem |